RAKESH K

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Profile Summary

Computer Science and Engineering student with hands-on experience in Java, React, Python, Flask, SQL, and REST API development through full-stack academic projects. Strong foundation in OOP, DBMS, and software engineering principles. Passionate about cloud applications, integration technologies, and enterprise software development.

Skills

Programming Languages: Java, Python, C, JavaScript
Frameworks & Technologies: React, Next.js, Node.js, Flask, HTML5, CSS3, REST APIs
Databases: MySQL, MongoDB, Firebase Firestore
Developer Tools: Git, Git Bash, Docker, Kubernetes
Core Computer Science: Data Structures & Algorithms, OOP, DBMS

Education

B.E. Computer Science & Engineering, Alva's Institute of Engineering and Technology, Mangalore — 2023 – Present
CGPA: 7.6

Pre-University (Science), ASC PU College, Bangalore — 2021 – 2023
Percentage: 70%

Secondary Education, ST Xavier High School, Bangalore — 2018 – 2021
Percentage: 79.51%

Projects

Faculty Appraisal Portal *(Web Application)* — GitHub

- Developed a full-stack MERN (MongoDB, Express.js, React, Node.js) application automating faculty self-appraisal workflows across four roles (Admin, Faculty, HOD, Principal).
- Implemented JWT authentication, RBAC, and bcrypt hashing to secure REST APIs and protect appraisal data.
- Built automated email verification, Multer file uploads, and Puppeteer-based PDF generation for appraisal proofs and reports.

Vaccination Management System *(Full Stack)* — GitHub

- Built a full-stack pediatric vaccination platform using Next.js 15, TypeScript, and Firebase Firestore for dual-schedule (UIP/IAP) tracking.
- Implemented atomic Firestore batch writes to ensure data consistency across vaccination records, appointment status, and inventory deduction.
- Integrated Google Genkit with Gemini 2.5 Flash to power an AI-driven Smart Availability assistant using Zod-validated structured outputs.

Target-X: CRISPR Target Analysis Platform *(AI / Full Stack)* — GitHub

- Developed an AI-powered platform using React, Flask, and Machine Learning to automate CRISPR target analysis.
- Designed a gRNA ranking engine evaluating guide RNAs using on-target efficiency and off-target risk metrics.
- Built an interactive React dashboard integrating REST APIs for real-time genome editing visualization.

Achievements

- Shortlisted among the Top 30 teams out of over 100 participating teams at Udgama National Level Hackathon 2025, Mysore.
- Advanced to the final round of the pan-India ICFAI GenAI Hackathon 2025 in Bangalore, rapidly prototyping an AI solution within a 24-hour build cycle as part of a 4-member team.

Certifications

- **Cloud Computing** – Successfully completed Cloud Computing NPTEL course
- **Data Analytics Job Simulation** – Deloitte, via Forage (July 2026), completed practical tasks in data analysis and forensic technology.
- **Fundamentals of Docker & Kubernetes** – Scaler Masterclass (July 2026), completed hands-on masterclass on containerization and orchestration.